Solution

1. (2×2 games). Find all pure and mixed Nash equilibriums in the following games.

Meeting in mountains. Two hiking groups (the Red and the Blue) choose where to cross the mountain ridge: through the Northern pass or through the Southern pass. They start from the different sides of the ridge and do not have any connection to each other. If they both choose the Northern pass, then they meet at the pass with probability 90%. If they both choose the Southern pass, then they meet with probability 70%. If they choose different passes, they do not meet at all. The Red get 10 units if they meet, and the Blue get 20 units in this case. For the Red, the cost of going through the Northern pass is 5 and the cost of going through the Southern pass is 7. For the Blue, the costs are 10 for the Northern pass and 8 for the Southern pass. Draw the matrices and solve the game. (The players are interested only in the expected profit.)

Solution.

Payoff matrix:

	Blue N	Blue S
Red N	$(10,20)$	$(0,0)$
Red S	$(0,0)$	$(10,20)$

after applying probabilities

$\downarrow$

	Blue N	Blue S
Red N	$(9,18)$	$(0,0)$
Red S	$(0,0)$	$(7,14)$

after applying the costs

$\downarrow$

	Blue N	Blue S
Red N	$(4,8)$	$(-5,-8)$
Red S	$(-7,-10)$	$(0,6)$

Pure Nash equilibriums.

$$(N,N), \qquad (S,S)$$

Mixed Nash equilibrium.

Let

$$p = \mathbb{P}(\text{Red plays } N), \qquad q = \mathbb{P}(\text{Blue plays } N)$$

Red indifferent.

Red's expected payoff from playing N is

$$u_R(N) = 4q + (-5)(1-q) = 9q - 5$$

Red's expected payoff from playing S is

$$u_R(S) = (-7)q + 0 \cdot (1-q) = -7q$$

$$9q - 5 = -7q$$

Hence

$$16q = 5, \qquad q = \frac{5}{16}$$

Blue indifferent.
Blue's expected payoff from playing N is

$$u_B(N) = 8p + (-10)(1-p) = 18p - 10$$

Blue's expected payoff from playing S is

$$u_B(S) = (-8)p + 6(1-p) = 6 - 14p$$

Set them equal:

$$18p - 10 = 6 - 14p$$

Hence

$$32p = 16, \qquad p = \frac{1}{2}$$

Therefore the mixed Nash equilibrium is

$$p = \frac{1}{2}, \qquad q = \frac{5}{16}$$

So, Red plays N with probability $\frac{1}{2}$ and Blue plays N with probability $\frac{5}{16}$
Penalty kicks. A Forward and a Goalkeeper play a penalty kick. The Forward can shoot to the left corner or to the right corner. If he shoots on the right, then he is always on target. But if he shoots on the left, then he is on target only with probability $\frac{5}{6}$. The Goalkeeper can jump to the left corner or to the right corner. If he jumps to the left and the ball is also on the left, then he always catches the ball. But if he jumps to the right, then he catches the ball only with probability $\frac{2}{3}$. If the ball and the Goalkeeper are in different corners, then the Goalkeeper never catches the ball. The Forward gets 6 points if he hits the target, -6 points if he misses the target, and 0 points if the Goalkeeper catches the ball. The Goalkeeper gets 6 points if he catches the ball, -6 points if the Forward hits the target, and 0 points if the Forward misses. Draw the matrices and solve the game. (The players are interested only in expected outcomes).
Solution.
Let the Forward (F) choose rows and the Goalkeeper (G) choose columns. Each player has two strategies:

$$L = \text{left corner}, \qquad R = \text{right corner}$$

Payoff matrix.

Case (L, L)**:**

$$u_F(L, L) = \frac{5}{6} \cdot 0 + \frac{1}{6} \cdot (-6) = -1,$$

$$u_G(L, L) = \frac{5}{6} \cdot 6 + \frac{1}{6} \cdot 0 = 5$$

Case (L, R)**:**

$$u_F(L, R) = \frac{5}{6} \cdot 6 + \frac{1}{6} \cdot (-6) = 4,$$

$$u_G(L, R) = \frac{5}{6} \cdot (-6) + \frac{1}{6} \cdot 0 = -5$$

Case (R, L)**:**

$$u_F(R, L) = 6,$$

$$u_G(R, L) = -6$$

Case (R, R)**:**

$$u_F(R, R) = \frac{2}{3} \cdot 0 + \frac{1}{3} \cdot 6 = 2,$$

$$u_G(R, R) = \frac{2}{3} \cdot 6 + \frac{1}{3} \cdot (-6) = 2$$

Hence the payoff matrix is

	G L	G R
F L	$(-1, 5)$	$(4, -5)$
F R	$(6, -6)$	$(2, 2)$

Pure Nash equilibriums.
Forward's best responses:

- If G plays L: $-1 < 6$, so F prefers R.
- If G plays R: $4 > 2$, so F prefers L.

Goalkeeper's best responses:

- If F plays L: $5 > -5$, so G prefers L.
- If F plays R: $2 > -6$, so G prefers R.

There is no pure Nash equilibrium.
Mixed Nash equilibrium.
Let

$$p = \mathbb{P}(\text{F plays } L), \qquad q = \mathbb{P}(\text{G plays } L)$$

Forward indifferent.

$$u_F(L) = q(-1) + (1 - q)4 = 4 - 5q,$$

$$u_F(R) = q \cdot 6 + (1 - q) \cdot 2 = 2 + 4q$$

$$4 - 5q = 2 + 4q \;\Rightarrow\; q = \frac{2}{9}$$

Goalkeeper indifferent.

$$u_G(L) = p \cdot 5 + (1-p)(-6) = 11p - 6,$$

$$u_G(R) = p(-5) + (1-p)2 = 2 - 7p$$

$$11p - 6 = 2 - 7p \;\Rightarrow\; p = \frac{4}{9}$$

That is,

$$\text{Forward plays } L \text{ with probability } \frac{4}{9}, \quad \text{Goalkeeper plays } L \text{ with probability } \frac{2}{9}$$

2. (Parametric game 2×3). Consider the following game with parameter a:

	L	M	R
U	$(4,1)$	$(3,6)$	$(a,7)$
D	$(5,6)$	$(2,5)$	$(5,3)$

where the row player chooses U, D and the column player chooses L, M, R. Find all pure and mixed Nash equilibriums in the cases:

(a) $a = 3$;

(b) $a = 6$;

(c) $a = 5$.

Solution.
We solve the game for $a = 3$, $a = 6$, and $a = 5$.
Step 1: Best responses in general.
For the row player:

- Against L: $4 < 5$, so the best response is D
- Against M: $3 > 2$, so the best response is U
- Against R: compare a and 5:

$$\begin{cases} D, & a < 5, \\ U \text{ and } D, & a = 5, \\ U, & a > 5 \end{cases}$$

For the column player:

- Against U: compare $1, 6, 7$, so the best response is R
- Against D: compare $6, 5, 3$, so the best response is L

Thus:

- (D, L) is always a pure Nash equilibrium
- (U, R) is a pure Nash equilibrium if $a \geq 5$

Step 2: Mixed equilibrium conditions.
Let the row player choose U with probability p, and the column player choose

$$L, M, R$$

with probabilities

$$x, y, z, \qquad x + y + z = 1$$

For the row player to mix, he must be indifferent between U and D:

$$4x + 3y + az = 5x + 2y + 5z$$

Hence

$$y = x + (5 - a)z \tag{1}$$

For the column player, expected payoffs are:

$$u_C(L) = p \cdot 1 + (1 - p) \cdot 6 = 6 - 5p,$$

$$u_C(M) = p \cdot 6 + (1 - p) \cdot 5 = 5 + p,$$

$$u_C(R) = p \cdot 7 + (1 - p) \cdot 3 = 3 + 4p$$

Equality conditions:

$$u_C(L) = u_C(M) \iff 6 - 5p = 5 + p \iff p = \frac{1}{6},$$

$$u_C(M) = u_C(R) \iff 5 + p = 3 + 4p \iff p = \frac{2}{3},$$

$$u_C(L) = u_C(R) \iff 6 - 5p = 3 + 4p \iff p = \frac{1}{3}$$

We now analyze each case separately.
(a) Case $a = 3$.
The matrix is

	L	M	R
U	$(4, 1)$	$(3, 6)$	$(3, 7)$
D	$(5, 6)$	$(2, 5)$	$(5, 3)$

Pure Nash equilibriums.
Since $a = 3 < 5$, against R the row player prefers D. Therefore (U, R) is not a Nash equilibrium.
So the only pure Nash equilibrium is

$$(D, L)$$

Mixed Nash equilibriums.
For $a = 3$, equation (1) becomes

$$y = x + 2z. \tag{2}$$

Mixing between L and M. This requires $p = \frac{1}{6}$, and then L and M are optimal for the column player.
Set $z = 0$. Then from (2),

$$y = x$$

Together with $x + y = 1$, we get

$$x = y = \frac{1}{2}$$

Hence one mixed equilibrium is:

$$p = \frac{1}{6}, \qquad (x, y, z) = \left(\frac{1}{2}, \frac{1}{2}, 0\right)$$

Mixing between M and R. This requires $p = \frac{2}{3}$, and then M and R are optimal for the column player.
Set $x = 0$. Then from (2),

$$y = 2z$$

Together with $y + z = 1$, we get

$$z = \frac{1}{3}, \qquad y = \frac{2}{3}$$

Hence another mixed equilibrium is:

$$p = \frac{2}{3}, \qquad (x, y, z) = \left(0, \frac{2}{3}, \frac{1}{3}\right)$$

Mixing between L and R. This would require $p = \frac{1}{3}$. But then

$$u_C(M) = 5 + \frac{1}{3} = \frac{16}{3}, \qquad u_C(L) = u_C(R) = \frac{13}{3},$$

so M is strictly better. Thus no equilibrium of this type exists.
Answer for $a = 3$.
Pure Nash equilibrium:

$$(D, L)$$

Mixed Nash equilibriums:

$$\left(\frac{1}{6}U + \frac{5}{6}D,\ \frac{1}{2}L + \frac{1}{2}M\right),$$

$$\left(\frac{2}{3}U + \frac{1}{3}D,\ \frac{2}{3}M + \frac{1}{3}R\right)$$

(b) Case $a = 6$.
The matrix is

	L	M	R
U	$(4, 1)$	$(3, 6)$	$(6, 7)$
D	$(5, 6)$	$(2, 5)$	$(5, 3)$

Pure Nash equilibriums.
Since $a = 6 > 5$, against R the row player prefers U. Hence (U, R) is a Nash equilibrium.
Therefore the pure Nash equilibriums are

$$(D, L), \qquad (U, R)$$

Mixed Nash equilibriums.
For $a = 6$, equation (1) becomes

$$y = x - z. \tag{3}$$

Equivalently,

$$x = y + z$$

Using $x + y + z = 1$, we obtain

$$(y + z) + y + z = 1 \quad \Rightarrow \quad 2y + 2z = 1 \quad \Rightarrow \quad x = \frac{1}{2}, \qquad y + z = \frac{1}{2}$$

Mixing between L and M. This requires $p = \frac{1}{6}$. Set $z = 0$. Then

$$x = \frac{1}{2}, \qquad y = \frac{1}{2}$$

So we get the mixed equilibrium

$$p = \frac{1}{6}, \qquad (x, y, z) = \left(\frac{1}{2}, \frac{1}{2}, 0\right)$$

Mixing between M and R. This would require $p = \frac{2}{3}$, but such a mixture has $x = 0$, contradicting $x = \frac{1}{2}$. So no such equilibrium exists.
Mixing between L and R. This would require $p = \frac{1}{3}$, but then M is strictly better than both L and R. So no such equilibrium exists.
Answer for $a = 6$.
Pure Nash equilibriums:

$$(D, L), \qquad (U, R)$$

Mixed Nash equilibrium:

$$\left(\frac{1}{6}U + \frac{5}{6}D,\ \frac{1}{2}L + \frac{1}{2}M\right)$$

(c) Case $a = 5$.

The matrix is

	L	M	R
U	$(4,1)$	$(3,6)$	$(5,7)$
D	$(5,6)$	$(2,5)$	$(5,3)$

Pure Nash equilibriums.
Since $a = 5$, against R the row player is indifferent between U and D. Therefore (U, R) is a Nash equilibrium, and as before (D, L) is also a Nash equilibrium. Thus the pure Nash equilibriums are

$$(D, L), \qquad (U, R)$$

Mixed Nash equilibriums.
For $a = 5$, equation (1) becomes

$$y = x. \tag{4}$$

Mixing between L and M. This requires $p = \frac{1}{6}$. Set $z = 0$. Then by (4),

$$x = y$$

Since $x + y = 1$, we get

$$x = y = \frac{1}{2}$$

Hence one mixed equilibrium is

$$p = \frac{1}{6}, \qquad (x, y, z) = \left(\frac{1}{2}, \frac{1}{2}, 0\right)$$

Column player uses only R. Suppose the column player plays R with probability 1, i.e.

$$(x, y, z) = (0, 0, 1)$$

Against R, the row player gets

$$u_R(U) = 5, \qquad u_R(D) = 5,$$

so any mixture of U and D is a best response.
Now we need R to be a best response for the column player. This means

$$u_C(R) \geq u_C(L), \qquad u_C(R) \geq u_C(M)$$

That is,

$$3 + 4p \geq 6 - 5p, \qquad 3 + 4p \geq 5 + p$$

These inequalities give

$$9p \geq 3 \iff p \geq \frac{1}{3}, \qquad 3p \geq 2 \iff p \geq \frac{2}{3}$$

Hence the condition is

$$p \geq \frac{2}{3}$$

Therefore, for every

$$p \in \left[\frac{2}{3}, 1\right],$$

the profile

$$(pU + (1-p)D,\ R)$$

is a Nash equilibrium.
Mixing between M and R. If $x = 0$, then from (4) also $y = 0$, so $z = 1$. This is exactly the previous family with pure R; there is no additional interior mixture of M and R.
Mixing between L and R. This would require $p = \frac{1}{3}$, but then M is strictly better than both L and R, so no such equilibrium exists.
Answer for $a = 5$.
Pure Nash equilibriums:

$$(D, L), \qquad (U, R)$$

Additional mixed equilibriums:

$$\left(\frac{1}{6}U + \frac{5}{6}D,\ \frac{1}{2}L + \frac{1}{2}M\right),$$

and the whole family

$$(pU + (1-p)D,\ R), \qquad p \in \left[\frac{2}{3}, 1\right]$$

Final answers.

$$\text{For } a = 3: \ (D, L),\ \left(\frac{1}{6}U + \frac{5}{6}D, \frac{1}{2}L + \frac{1}{2}M\right),\ \left(\frac{2}{3}U + \frac{1}{3}D, \frac{2}{3}M + \frac{1}{3}R\right)$$

$$\text{For } a = 6: \ (D, L),\ (U, R),\ \left(\frac{1}{6}U + \frac{5}{6}D, \frac{1}{2}L + \frac{1}{2}M\right)$$

For $a = 5: \ (D, L),\ (U, R),\ \left(\frac{1}{6}U + \frac{5}{6}D, \frac{1}{2}L + \frac{1}{2}M\right),\ (pU + (1-p)D,\ R),\ p \in \left[\frac{2}{3}, 1\right]$

3. (Abstract game). Consider the following game:

	t_1	t_2	t_3	t_4	t_5
s_1	$(2,0)$	$(5,3)$	$(2,4)$	$(6,1)$	$(3,2)$
s_2	$(3,1)$	$(0,6)$	$(0,5)$	$(3,2)$	$(0,3)$
s_3	$(4,6)$	$(1,3)$	$(2,1)$	$(5,4)$	$(1,3)$
s_4	$(2,7)$	$(4,2)$	$(1,4)$	$(5,8)$	$(4,5)$

where the row player chooses s_1, s_2, s_3, s_4 and the column player chooses t_1, t_2, t_3, t_4, t_5.

(a) By eliminating strictly dominated strategies reduce the game to the form $2 \times N$. Specify which strategy is dominated by which one. Note that a strategy may be dominated not only by a particular strategy but also by a mixture of strategies. Any further eliminations must be impossible in the remaining game.

(b) Find all pure and mixed equilibriums in the remaining $2 \times N$ game.

Solution.
(a) Elimination of strictly dominated strategies.
Step 1: eliminate s_2. Compare the row-player payoffs of s_2 and s_3:

$$(3, 0, 0, 3, 0) \quad \text{vs.} \quad (4, 1, 2, 5, 1)$$

Since

$$4 > 3, \quad 1 > 0, \quad 2 > 0, \quad 5 > 3, \quad 1 > 0,$$

strategy s_3 strictly dominates s_2. Hence s_2 is eliminated.
The reduced game is

	t_1	t_2	t_3	t_4	t_5
s_1	$(2,0)$	$(5,3)$	$(2,4)$	$(6,1)$	$(3,2)$
s_3	$(4,6)$	$(1,3)$	$(2,1)$	$(5,4)$	$(1,3)$
s_4	$(2,7)$	$(4,2)$	$(1,4)$	$(5,8)$	$(4,5)$

Step 2: eliminate t_5.
We want to show that column t_5 is strictly dominated by a mixture of other columns. Let the mixture of columns t_1, t_2, t_3, t_4 have weights

$$\alpha, \beta, \gamma, \delta \geq 0, \qquad \alpha + \beta + \gamma + \delta = 1$$

Now calculate the expected payoffs of the mixture.
We only need to check the remaining rows s_1, s_3, s_4. The column player's expected payoffs are:

- Against s_1:
$$0\alpha + 3\beta + 4\gamma + 1\delta$$
- Against s_3:
$$6\alpha + 3\beta + 1\gamma + 4\delta$$
- Against s_4:
$$7\alpha + 2\beta + 4\gamma + 8\delta$$

Compare with t_5.
Column t_5 yields payoffs

$$(2, 3, 5)$$

To strictly dominate t_5, the mixture must satisfy:

$$3\beta + 4\gamma + \delta > 2,$$

$$6\alpha + 3\beta + \gamma + 4\delta > 3,$$

$$7\alpha + 2\beta + 4\gamma + 8\delta > 5,$$

together with

$$\alpha + \beta + \gamma + \delta = 1, \qquad \alpha, \beta, \gamma, \delta \geq 0$$

Thus we obtain a system of linear inequalities. Any solution gives a valid dominating mixture.

Identify strong columns.

Compare each column to $t_5 = (2, 3, 5)$:

- For s_1: need > 2, good columns are $t_2 = 3$, $t_3 = 4$
- For s_3: need > 3, good columns are $t_1 = 6$, $t_4 = 4$
- For s_4: need > 5, good columns are $t_1 = 7$, $t_4 = 8$

Hence we try a mixture of t_2, t_3, t_4:

$$\frac{1}{3}t_2 + \frac{1}{6}t_3 + \frac{1}{2}t_4$$

Its column-player payoffs against s_1, s_3, s_4 are

$$\left(\frac{1}{3} \cdot 3 + \frac{1}{6} \cdot 4 + \frac{1}{2} \cdot 1,\ \frac{1}{3} \cdot 3 + \frac{1}{6} \cdot 1 + \frac{1}{2} \cdot 4,\ \frac{1}{3} \cdot 2 + \frac{1}{6} \cdot 4 + \frac{1}{2} \cdot 8\right) = \left(\frac{13}{6}, \frac{19}{6}, \frac{16}{3}\right)$$

For t_5, the column-player payoffs are

$$(2, 3, 5)$$

Since

$$\frac{13}{6} > 2, \qquad \frac{19}{6} > 3, \qquad \frac{16}{3} > 5,$$

strategy t_5 is strictly dominated by the mixture

$$\frac{1}{3}t_2 + \frac{1}{6}t_3 + \frac{1}{2}t_4$$

Hence t_5 is eliminated.

The reduced game is now

	t_1	t_2	t_3	t_4
s_1	$(2,0)$	$(5,3)$	$(2,4)$	$(6,1)$
s_3	$(4,6)$	$(1,3)$	$(2,1)$	$(5,4)$
s_4	$(2,7)$	$(4,2)$	$(1,4)$	$(5,8)$

Step 3: eliminate s_4. Consider the mixture

$$\frac{5}{6}s_1 + \frac{1}{6}s_3$$

Its row-player payoffs against t_1, t_2, t_3, t_4 are

$$\left(\frac{5}{6}\cdot 2+\frac{1}{6}\cdot 4,\ \frac{5}{6}\cdot 5+\frac{1}{6}\cdot 1,\ \frac{5}{6}\cdot 2+\frac{1}{6}\cdot 2,\ \frac{5}{6}\cdot 6+\frac{1}{6}\cdot 5\right)=\left(\frac{7}{3},\frac{13}{3},2,\frac{35}{6}\right)$$

For s_4, the row-player payoffs are

$$(2,4,1,5)$$

Since

$$\frac{7}{3}>2, \qquad \frac{13}{3}>4, \qquad 2>1, \qquad \frac{35}{6}>5,$$

strategy s_4 is strictly dominated by the mixture

$$\frac{5}{6}s_1+\frac{1}{6}s_3$$

Hence s_4 is eliminated.
The reduced game is now

	t_1	t_2	t_3	t_4
s_1	$(2,0)$	$(5,3)$	$(2,4)$	$(6,1)$
s_3	$(4,6)$	$(1,3)$	$(2,1)$	$(5,4)$

Step 4: eliminate t_4. Consider the mixture

$$\frac{2}{3}t_1+\frac{1}{3}t_3$$

Its column-player payoffs against s_1, s_3 are

$$\left(\frac{2}{3}\cdot 0+\frac{1}{3}\cdot 4,\ \frac{2}{3}\cdot 6+\frac{1}{3}\cdot 1\right)=\left(\frac{4}{3},\frac{13}{3}\right)$$

For t_4, the column-player payoffs are

$$(1,4)$$

Since

$$\frac{4}{3}>1, \qquad \frac{13}{3}>4,$$

strategy t_4 is strictly dominated by the mixture

$$\frac{2}{3}t_1+\frac{1}{3}t_3$$

Hence t_4 is eliminated.
We arrive at the 2×3 game

	t_1	t_2	t_3
s_1	$(2,0)$	$(5,3)$	$(2,4)$
s_3	$(4,6)$	$(1,3)$	$(2,1)$

No further eliminations are possible. For the row player, neither s_1 nor s_3 strictly dominates the other:

$$2 < 4 \text{ at } t_1, \qquad 5 > 1 \text{ at } t_2$$

For the column player, none of t_1, t_2, t_3 is strictly dominated in the remaining game:

- t_1 gives the highest payoff against s_3,
- t_3 gives the highest payoff against s_1,
- t_2 is not dominated since it will be a best response for some mixed strategy of the row player (see part (b)).

(b) Pure and mixed Nash equilibriums in the remaining 2×3 game.
We now analyze

	t_1	t_2	t_3
s_1	$(2,0)$	$(5,3)$	$(2,4)$
s_3	$(4,6)$	$(1,3)$	$(2,1)$

Pure Nash equilibriums.
Row player's best responses.

- Against t_1: $2 < 4$, so the best response is s_3.
- Against t_2: $5 > 1$, so the best response is s_1.
- Against t_3: $2 = 2$, so both s_1 and s_3 are best responses.

Column player's best responses.

- Against s_1: compare $0, 3, 4$, so the best response is t_3.
- Against s_3: compare $6, 3, 1$, so the best response is t_1.

Therefore the pure Nash equilibriums are

$$\boxed{(s_3, t_1) \quad \text{and} \quad (s_1, t_3).}$$

Mixed Nash equilibriums.
Let the row player choose s_1 with probability p, and s_3 with probability $1 - p$. Then the column player's expected payoffs are

$$u_C(t_1) = 0 \cdot p + 6(1 - p) = 6 - 6p,$$
$$u_C(t_2) = 3,$$
$$u_C(t_3) = 4p + 1(1 - p) = 1 + 3p$$

Best responses of the column player. We compare:

$$u_C(t_1) = u_C(t_2) \iff 6 - 6p = 3 \iff p = \frac{1}{2},$$
$$u_C(t_2) = u_C(t_3) \iff 3 = 1 + 3p \iff p = \frac{2}{3}$$

Hence:

- if $p < \frac{1}{2}$, the unique best response is t_1,
- if $p = \frac{1}{2}$, the best responses are t_1, t_2,
- if $\frac{1}{2} < p < \frac{2}{3}$, the unique best response is t_2,
- if $p = \frac{2}{3}$, the best responses are t_2, t_3,
- if $p > \frac{2}{3}$, the unique best response is t_3.

A fully mixed row player. Let the column player choose t_1, t_2, t_3 with probabilities x, y, z, where

$$x + y + z = 1$$

The row player's expected payoffs are

$$u_R(s_1) = 2x + 5y + 2z,$$

$$u_R(s_3) = 4x + y + 2z$$

For the row player to mix, we need indifference:

$$2x + 5y + 2z = 4x + y + 2z$$

which simplifies to

$$4y = 2x \iff x = 2y. \tag{1}$$

Mixed equilibrium with support $\{t_1, t_2\}$. At $p = \frac{1}{2}$, the column player's best responses are t_1, t_2. So set $z = 0$. Then from

$$x + y = 1, \qquad x = 2y,$$

we obtain

$$y = \frac{1}{3}, \qquad x = \frac{2}{3}$$

Thus one mixed Nash equilibrium is

$$\left(\frac{1}{2}s_1 + \frac{1}{2}s_3,\ \frac{2}{3}t_1 + \frac{1}{3}t_2\right)$$

Nash equilibriums in which the column player chooses t_3. If the column player plays t_3 purely, then the row player's payoffs are

$$u_R(s_1) = 2, \qquad u_R(s_3) = 2,$$

so any mixture of s_1 and s_3 is a best response.
For t_3 to be a best response for the column player, we need

$$u_C(t_3) \geq u_C(t_1), \qquad u_C(t_3) \geq u_C(t_2),$$

that is,

$$1 + 3p \geq 6 - 6p, \qquad 1 + 3p \geq 3$$

$$9p \geq 5 \iff p \geq \frac{5}{9}, \qquad 3p \geq 2 \iff p \geq \frac{2}{3}$$

So the condition is

$$p \geq \frac{2}{3}$$